Soil Detection Model and Application Based on Hyperspectral Image Classification Using the Gia-CFSL Model

Abstract

This study addresses the limitations of traditional soil testing and hyperspectral image classification techniques by developing a soil detection model based on the Gia-CFSL framework. Through parameter optimization and various training strategies, the model was validated across multiple soil datasets. Results show that compared to conventional methods and other deep learning models, this approach achieves higher accuracy and stronger generalization capability in detecting soil carbon content and elemental composition, offering a new pathway for soil analysis.

Keywords: hyperspectral image; soil classification; Gia-CFSL model

1 Introduction

1.1 Research Background and Significance

Soil is a critical resource for agriculture, ecology, and environmental management, with its quality directly affecting crop growth, land use efficiency, and ecosystem health. Traditional soil testing relies on laboratory analysis, which, although accurate, is time-consuming, labor-intensive, and limited in sample representativeness. Hyperspectral image classification technology, by capturing multi-band reflectance data, enables efficient, precise, and non-invasive analysis of soil properties, overcoming the limitations of conventional methods and offering a new solution for large-scale soil monitoring.

However, the practical application of hyperspectral imaging technology is often limited by the scarcity of samples. Traditional machine learning methods require large amounts of labeled data, but obtaining extensive, high-quality labeled soil samples in real-world scenarios typically involves substantial manual effort, making it time-consuming and costly. To address this challenge, the Gia-CFSL (Graph Information Aggregation Cross-Domain Few-Shot Learning) model integrates few-shot learning with cross-domain adaptation techniques, enabling highly accurate soil classification and pollution monitoring with only a small number of labeled samples, significantly improving classification performance under low-sample conditions. This technology offers an efficient and low-cost new approach to soil detection, holding significant implications for precision agriculture, environmental monitoring, and sustainable development.

1.2 Current Research Status at Home and Abroad

1.2.1 Current Research Status in China

Currently, research in China on the application of hyperspectral image classification in soil detection mainly focuses on soil type classification and pollution monitoring. Technically, related studies primarily concentrate on improving hyperspectral image data preprocessing methods, enhancing algorithm models, and reducing reliance on

Large amounts of high-quality labeled samples are required, among other factors. Most existing research employs classification methods such as supervised learning algorithms, including support vector machines[3].
Support Vector Machines (SVM), Random Forests (RF), Neural Networks (NN), etc.; unsupervised learning algorithms such as clustering analysis and k-means

algorithms, etc. In recent years, with the rapid development of deep learning, convolutional neural networks (CNNs) and long short-term memory (LSTM) networks [4]
These methods have also been applied to hyperspectral image classification, with continuous in-depth improvements made upon them. [5] [6][7], achieved good results,

It has shown outstanding performance, especially in the automatic identification of soil types and compositions. To address the issue of limited sample size, relevant studies have combined deep learning algorithms with methods such as dynamic convolution and dual-branch multi-scale convolution to progressively reduce the impact of sample quantity on detection results. In practical applications [8] [9]
Hyperspectral image classification technology has already been widely deployed on a large scale.

1.2.2 Current Research Status Abroad

Similar to domestic research, international studies have primarily focused on improving algorithm models, building upon machine learning or deep learning techniques and innovating further based on practical challenges [10]. In terms of research directions, foreign studies mainly concentrate on soil composition analysis [11], particularly the detection of organic matter, moisture, and heavy metals, for applications in agricultural production and environmental protection; soil type classification and mapping [12], applied in natural geography; and multi-source data fusion, such as integrating LiDAR topographic data with hyperspectral imagery, to enhance the accuracy of soil texture classification. While international research is more diversified and utilizes a broader range of training datasets, differences still exist compared to domestic studies due to variations in soil characteristics.

1.2.3 Limitations of the current research

Current research on few-shot learning capabilities and cross-domain generalization of models, both domestically and internationally, remains insufficient. Traditional machine learning algorithms rely heavily on large amounts of labeled data and exhibit poor adaptability to cross-domain problems, limiting their widespread application in practice. Although the emergence of deep learning methods—particularly the use of models such as 3D convolutional neural networks (3D-CNNs)—has improved model performance to some extent, limitations still exist regarding cross-domain generalization. As a result, currently widely used models are unable to achieve satisfactory classification across different scenarios, and their performance is also suboptimal in cases with scarce samples. To further improve the accuracy of hyperspectral image classification while reducing costs, continued in-depth research into few-shot learning and cross-domain generalization techniques is essential.

1.3 Research Objectives and Content

The research objective of this project is to utilize the Gia-CFSL (Graph Information Aggregation Cross-Domain Few-Shot Learning) model to achieve cross-scenario hyperspectral image-based soil composition detection and analysis from source to target domains using limited sample data. The output includes measurements of phosphorus (P), potassium (K), magnesium (Mg) content, and pH value in the soil, along with an assessment of soil health conditions at the dataset's location, enabling farmers to implement targeted soil management practices.

To address the domain shift and few-shot learning challenges encountered in traditional methods, this project focuses on the Gia-CFSL model. First, we investigate information propagation in graph neural networks, establish a framework, and construct feature graphs (FG) and distribution graphs (DG) for the Gia-CFSL model to model non-local spatial relationships and capture global correlations between land cover classes and soil composition across different scenes. Next, we study cross-domain graph alignment by leveraging graph optimal transport (GOT) to align graph and distribution-level features between the source domain (SD) and target domain (TD), and aggregate cross-domain attention at both feature and higher-order distribution levels to enhance inter-domain similarity learning. Finally, we adopt an episodic training paradigm from few-shot learning (FSL) to simulate low-sample scenarios and improve the model's generalization capability for novel classes. In subsequent experimental evaluations, using cross-scenario SD and TD data, our model outperforms traditional deep learning models in both handling domain shift and achieving detection accuracy, marking a breakthrough in integrating domain adaptation (DA) with few-shot learning (FSL).

1.4 Research Methods and Technical Approach

1.4.1 Research Methodology

This project employs a research methodology combining theory and experimentation, involving the study of relevant literature to gather information and explore potential solutions to research problems, as well as writing and running code to observe the results.

1.4.2 Technical Roadmap

(1) Data Preprocessing and Sample Generation

During the initial model development phase, four datasets were used in total. The training set (Chikusei dataset) and the test sets (HyperDataset, Hyperspectral benchmark dataset on soil moisture, and soil_element datasets) spanned different scenarios, facilitating evaluation of the model's cross-scenario classification accuracy.

1. Data Preprocessing

Read the hyperspectral image and its corresponding label file from the dataset. The image data is a three-dimensional matrix (height × width × number of bands), and the labels are a two-dimensional matrix (height × width). Standardize the data for each band (mean = 0, variance = 1) to eliminate dimensional differences.

Pad the image edges with zeros using the zeroPadding_3D function (with padding length equal to patch_length) to ensure that boundary pixels can also generate complete data blocks during subsequent neighborhood sampling. For each target pixel, extract the multi-band data from its surrounding neighborhood of size (2*patch_length+1) × (2*patch_length+1), forming a local spatial-spectral joint feature block.

2. Sample Generation

Randomly select samples based on label distribution to ensure representation from each class. Generate a list named whole_indices containing the linear indices of all sampled pixels. Finally, convert the linear indices into 2D coordinates and extract neighborhood data blocks based on these coordinates.

3. Data Storage

Store the processed image data and labels in a dictionary, then serialize and save it.

(2) Building the model framework

This paper proposes the Gia-CFSL (Graph Information Aggregation Cross-Domain Few-Shot Learning) framework, whose core lies in enabling knowledge transfer from the source domain (SD) to the target domain (TD) through cross-domain graph alignment and cross-domain attention, addressing domain shift and few-shot challenges in hyperspectral classification across different scenes. The framework consists of the following modules:

1) Feature extraction network: A 3D residual CNN is used to extract spatial-spectral features (D_Res_3d_CNN class).

2) Dual-image generation module:

Feature Map (FG): Construct sample-level similarity relationships based on Euclidean distance (euclidean_metric function)

Distribution Graph (DG): Generates category-level distribution relationships through IDE-blocks (DPGN class).

3) Cross-domain Alignment Module:

Optimal Transport (OT): Align feature maps using the IPOT algorithm (IPOT_torch_batch);

Graph Optimal Transport (GOT): Align distribution graphs using the GW distance (GW_distance function).

4) Attention Aggregation Module: Computes inter-domain feature-level and distribution-level attention (AttentionalGNN type).

1. Feature Extraction and Graph Construction

1) Multi-scale feature extraction

Input the hyperspectral cube $\chi \in R^{H\times W\times B}$, extract features through a three-level structure: first extracting residual blocks (residual_block) to max pooling, then to secondary residual blocks and 3D convolution, outputting two hierarchical features: intermediate-layer feature $F\boxtimes \in R^{d\boxtimes}$ *and* deep-layer feature $F\boxtimes \in R^{d\boxtimes}$.

2) Generate dual images

Feature map: Compute the cosine distance matrix between the support set and the query set.

$$C_{ij} = 1 - \frac{f_i^T f_j}{\|f_i\| \|f_j\|} \quad (式\ 1-1)$$

Distribution map: Propagate F1 and F2 through the IDE-block to generate high-order statistical features.

2. Cross-domain Alignment Algorithm

1) Multi-feature Level Alignment (IPOT)

Minimize the Wasserstein distance between the source and target domains:

$$L_{OT} = \min_{T} \langle T, C \rangle_F + \beta H(T) \quad （式 1- 2）$$

where T is the transport matrix optimized via Sinkhorn iterations (located in the IPOT_torch_batch function).

2) Graph Structure Alignment (GOT)

Define GW distance loss:

$$\mathrm{L_{GW}} = \sum_{i,j,k,l} (C_s^{ik} - C_t^{jl})^2 \gamma_{ij} \gamma_{kl} \quad （式 1- 3）$$

where γ is the graph transfer matrix obtained by solving IPOT (located in the GW_torch_batch function).

3. Small Sample Classification Optimization

1) Original Task Training

Each episode samples a support set (S) and a query set (Q) for the task class from the source domain and target domain, respectively. Prototype computation:

$$P_c = \frac{1}{|S_c|} \sum_{f_i \in S_c} f_i \quad （式 1- 4）$$

2) Joint loss function

$$L = L_{CE}(Q_{src}) + L_{CE}(Q_{tar}) + \lambda_1 (L_{OT} + L_{GW}) + \lambda_2 L_{ATT} \quad （式 1- 5）$$

The small sample loss is $L_{CE(Qsrc)} + L_{CE(Qtar)}$, the alignment loss is $LOT + L_{GW}$, and the $_{attention}$ loss is LATT. $_{Among\ them,\ LATT}$ is computed by the CSA_{block} (cross_att_loss).

4. Key Code Correspondence Explanation

Model Module	Code implementation	Function Description
Feature extraction network	D_Res_3d_CNN	Spatial-spectral feature extraction
IDE-block	DPGN	Generate distribution maps and optimize GW distance
CSA-block	AttentionalGNN	Compute cross-domain feature/distribution attention
IPOT Optimization	IPOT_torch_batch	Solving the optimal transport matrix
Original task construction	Task+get_HBKC_data_loader	Small sample task sampling and data loading

Table 1-1 Key Code Description Diagram

2 Gia-CFSL Model Principles and Methods

2.1 Overview of Hyperspectral Image Classification

Hyperspectral remote sensing technology involves acquiring continuous spectral information of objects using fixed electromagnetic wave bands, thereby enabling the inversion of material composition and object identification based on the obtained spectral feature curves. Most materials exhibit diagnostic absorption characteristics within the solar reflectance spectrum range (300−2500 nm), and these spectral absorption features vary among different materials. This property allows for qualitative and quantitative analysis of material composition across multiple fields, facilitating the acquisition of ground-based information.

Compared to conventional RGB imaging, which uses three bands, and multispectral imaging, which typically employs up to a dozen bands, hyperspectral imaging can capture hundreds of spectral bands per spatial pixel, with the spectral channels often being continuous. This fine spectral resolution gives hyperspectral remote sensing a natural advantage in extracting spectral features from complex soil compositions. Given a dataset of hyperspectral images of soils with known levels of specific components, one can extract the multi-band reflectance information stored at each pixel and use deep learning to distinguish the reflectance characteristics across different concentration ranges of the component, thereby enabling classification and prediction of unknown soil samples' elemental content levels. At the current stage of technology, numerous academic studies have been conducted on retrieving soil moisture, elemental content, and heavy metal pollution using hyperspectral remote sensing. However, these studies employ diverse approaches in selecting spectral inversion models. While various machine learning algorithms offer increasingly sophisticated methods for classifying soil image pixels, most models still face challenges such as high complexity and relatively low accuracy.

2.2 Core Principles of the Gia-CFSL Model

In practical remote sensing applications, it is common for the available hyperspectral image data in a given research domain to be insufficient, thereby limiting the full potential of machine learning and deep learning models—designed for large datasets—to deliver optimal prediction or classification performance. However, hyperspectral image classification tasks across different domains share certain similarities: they all rely on identifying classification patterns by analyzing overall differences in reflectance values across multiple spectral bands among samples of different classes, as well as the similarity in hyperspectral reflectance characteristics among samples within the same class. Therefore, if we can leverage hyperspectral datasets from other well-labeled domains (source domains) and combine them with the limited labeled samples from our target domain (target domain), learn shared knowledge between the source and target domains, and transfer the model trained on the source domain to the target domain, we are more likely to achieve improved classification accuracy.

The Gia-CFSL model employs a meta-learning-based few-shot domain adaptation strategy, simultaneously capturing non-local relationships at both the graph feature and graph distribution levels from source and target training sets. By integrating graph optimal transport with attention-based feature similarity, it achieves cross-domain alignment, mitigating the impact of domain shift between source and target domains, thereby enabling a domain adaptation method suitable for scenarios where source and target domains involve different sensors and land cover classification tasks.

2.2.1 Cross-domain Few-shot Learning (FSL)

Few-shot learning enables rapid learning of new tasks with a very limited number of labeled examples. Each few-shot learning task T consists of a support set S and a query set Q, used respectively for building a classifier and evaluating its performance. Training is performed over episodes on the entire dataset, where each episode constructs a meta-task: the support set contains C classes, each with K samples—this meta-task is referred to as a C-way K-shot task—and the query set includes the same number of classes, with the number of samples per class set arbitrarily to T.

Construct meta-tasks *Ts and Tt for the source and* target domains respectively, *ensuring* they are greater than . Below, we illustrate the training process using as an example. From the source domain dataset, select classes, and randomly pick samples from each class to form the support set and query set. The model is trained on the support set, measuring the similarity between query set samples and support set prototypes (the average feature vector of K samples per class) by summing the Euclidean distances (ED) across all features, thereby obtaining the cross-entropy prediction loss.

$$P(\hat{y}_m^s|x_m^s) = \text{Softmax}\left(-\text{ED}\left(z_m^s, z_n^{\text{mean}(c)}\right)\right) \quad \text{（式 2- 1）}$$

$$\mathcal{L}_{CE}^{s}(P(\hat{y}_m^s|x_m^s), y_m^s) = -\, y_m^s \log \quad P(\hat{y}_m^s|x_m^s). \quad \text{（式 2- 2）}$$

[illegible]

Request a sample; when the amount of data for each class in the target dataset is insufficient, use data augmentation techniques to expand the data.

2.2.2 Graph Neural Networks

The GNN IDE is a dual-graph collaborative learning framework designed for cross-domain few-shot learning. By iteratively interacting between the point graph (modeling sample-level relationships) and the distribution graph (modeling class-level distributions), it aggregates intra-domain sample relationships, clusters features of similar samples, and reduces intra-class variations within domains, thereby achieving feature enhancement and domain adaptation. The analysis still relies on the meta-dataset, where the construction methods for point graphs and distribution graphs are consistent across source and target datasets, with all samples from both support and query sets merged to form the graphs.

(1) Click on the FG (Feature Graph Construction)

The dot plot is used to capture feature similarities among samples. Its nodes are:

$$v_{f(i)}^{s0} = f_{emb}(M_s(x_i^s)) \quad \text{（式 2- 3）}$$

[illegible]

where is the node of the i-th sample, and is the size of the feature-level information, i.e., the dimensionality of the embedding features contained in the i-th node. Edges in the graph are constructed based on feature similarity between pairs of nodes (samples):

$$e_{f(i,j)}^{s0} = f_{FG}\left(\left(v_{f(i)}^{s0} - v_{f(j)}^{s0}\right)^2\right) \quad \text{（式 2- 4）}$$

[illegible]距离平方，得到大小为的差异向量，再

(2) Distribution Map DG (Distribution Graph Construction)

The distribution graph models the class-level probabilistic distribution differences among samples and records the distributional distances of each sample relative to class prototypes. To capture intra-domain distributional information, the similarities between FG edge features are aggregated into node features for DG. Edge features related to the support set are extracted to measure the distance between each sample and the prototypes of respective classes, which are then combined with the originally initialized DG node features and dimensionally reduced to update the distribution graph node values.

$$v_{d(i)}^{s1} \leftarrow \text{MLP}([e_{f(i,j)}^{s0}, v_{d(i)}^{s0}]) \quad \text{（式 2- 5）}$$

[illegible]

DG node values are initialized differently between the support set and the query set. In the support set, sample class labels are converted into one-hot encoding. The adjacency matrix in vector form; initialize all node values in the query set to zero.

The edge weights in the distribution map are set similarly to those in the point graph.

(3) Point graph node aggregation update

$$v_{f(i)}^{sl} \leftarrow \text{MLP}\left(\left[\left(\sum_j e_{d(i,j)}^{sl} v_{p(j)}^{s0}\right), v_{f(i)}^{s0}\right]\right) \quad \text{（式 2- 6）}$$

Re-aggregate and update the feature nodes of FG using DG. The updated node features are then used to compute class prototypes (mean) in the meta-task, thereby incorporating non-local relationships into the meta-training process and mitigating domain shifts caused by differences between source and target domains, such as domain and environmental variations.

2.2.3 Optimal transmission

Optimal transport aims to find the minimum-cost scheme for transforming one probability distribution μ into another ν. This paper achieves cross-domain alignment through graph optimal transport in the model, employing two optimal transport distances: WD for node matching and GWD for edge matching. Four types of matching are computed separately between the source domain and target domain—between FG point graphs and DG distribution graphs—and a shared transportation plan is used to realize graph optimal transport (GOT), with the loss calculated accordingly.

$$\mathrm{GOT}(G^s, G') = \sum_{i,\hat{i},j,\hat{j}} (T_{ij} \cdot c(v_i^s, v_j') + T_{ij} \cdot L(v_i^s, v_j^t, \hat{v}_i^s, \hat{v}_j^t)) \quad \text{（式 2- 7）}$$

$$L_{\mathrm{GOT}} = \mathrm{GOT}(G_f^s, G_f^s) + \mathrm{GOT}(G_d^s, G_d^s). \text{（式 2- 8）}$$

The Cross-Domain Attention Generation Mechanism (CSA) aims to dynamically capture the correlations between source and target domains through attention mechanisms. By leveraging attention features, it coordinates feature similarity with high-order distributional attention similarity to form a quantifiable loss, enabling effective feature alignment and knowledge transfer under domain differences. This addresses the limitation of GOT, which lacks cross-domain information interaction and thus suffers from reduced domain shift mitigation.

The final loss function obtained for the entire model, used to measure performance and update parameters, is:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{CE}}^{\text{s}} + \mathcal{L}_{\text{CE}}^{\text{t}} + \lambda_1 \mathcal{L}_{\text{GOT}} + \lambda_2 \mathcal{L}_{\text{MMD}} \text{（式 2- 9）}$$

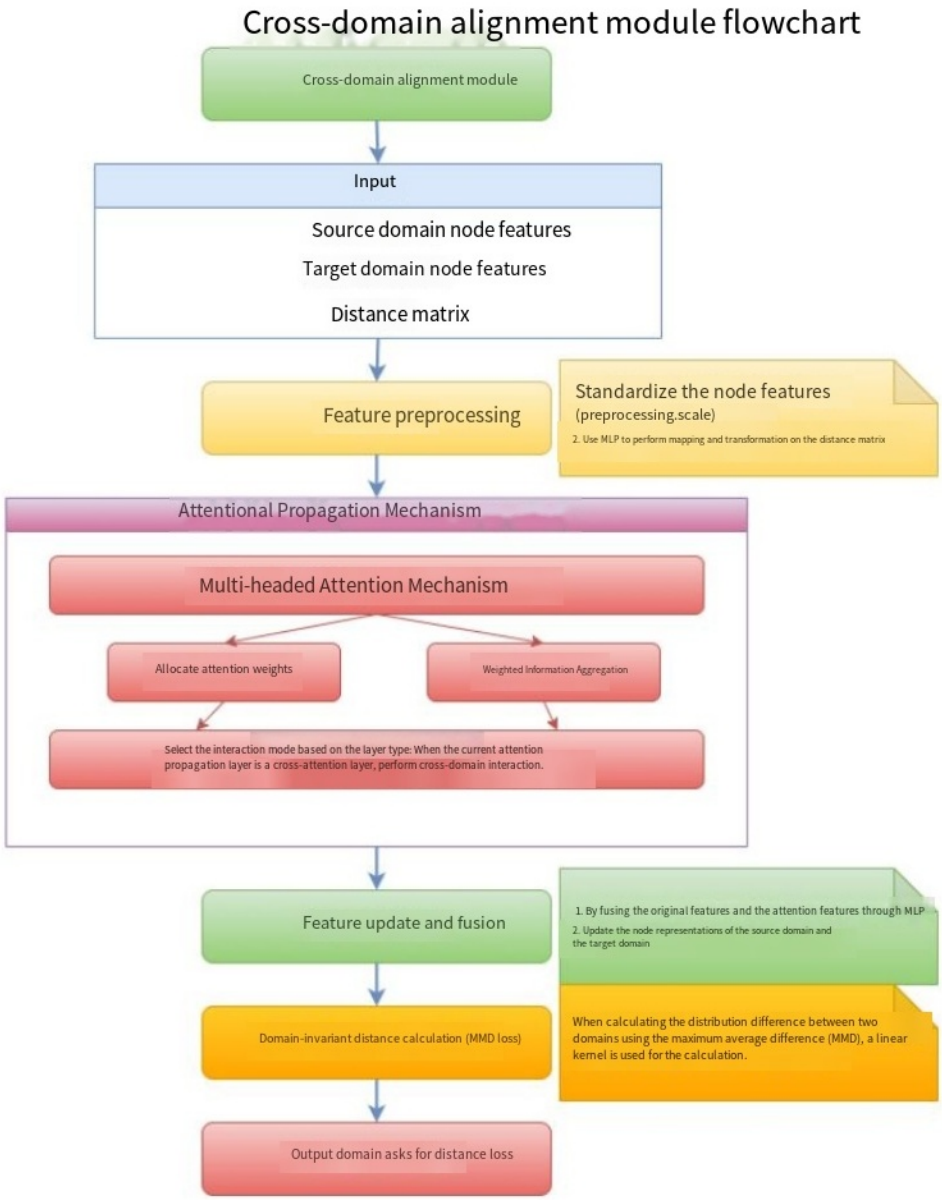


Figure 2-1 Cross-domain Alignment Module Flowchart

2.3 Model Building Steps

(1) Data Preprocessing

Load the source and target data, reorganize and convert the data formats so that each sample becomes a 9×9 pixel image, with each pixel containing multi-band reflectance information. Each sample corresponds to a label, and after data augmentation, generate the source dataset SD and target dataset TD suitable for meta-training, which are then fed into a meta-task training episode.

(2) Feature Extraction

Construct meta-tasks separately for the source and target domains: use the number of labels in the target domain, C, as the number of classes in the meta-task, with one sample per class. Randomly select C labels from the multiple labels in the source domain. For each selected label, take one sample as the support set and T samples as the query set; similarly, for each label in the target domain, take one sample as the support set and T samples as the query set. The remaining samples are used as the validation set to evaluate the final performance of the prediction model.

Considering that the hyperspectral band lengths between SD and TD may differ, 3D residual convolution is used for feature extraction to unify the dimensions from (9×9×number of bands×number of samples) to (128×number of samples), extracting 128 embedded features per sample.

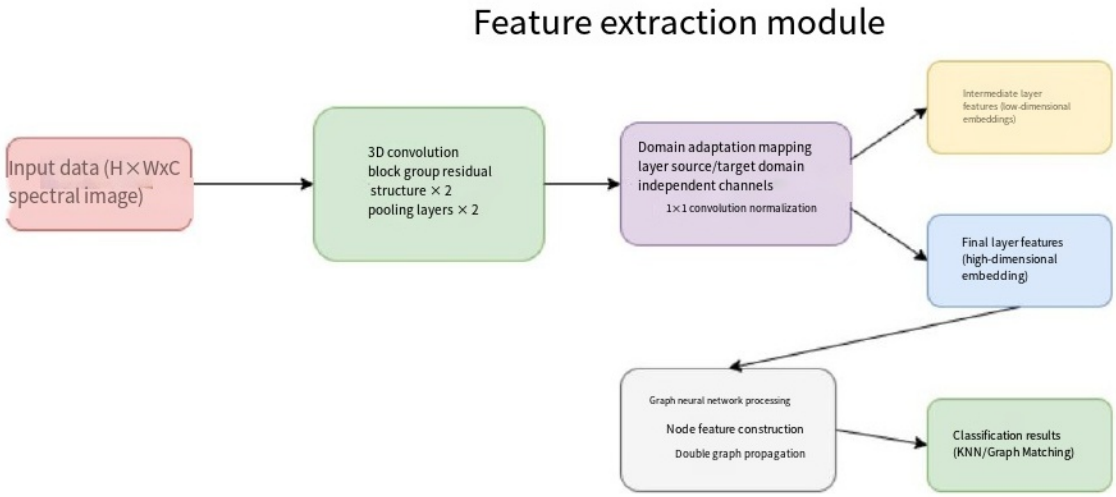


Figure 2-2 Feature Extraction Module

(3) Initialize feature map

Use the embedded features of support and query set samples from SD and TD as the initialization node information for the point graph; the edge information between the point graph and distribution graph is initialized in the same way, as described in the principle section.

(4) Perform IDE and CSA in sequence, compute the cross-entropy between the query set and category prototypes, and calculate the GOT loss. In IDE, construct point plots and distribution plots using C*1 + C*T samples.

(5) Backpropagation, updating the parameters of each model based on the overall model's loss function.

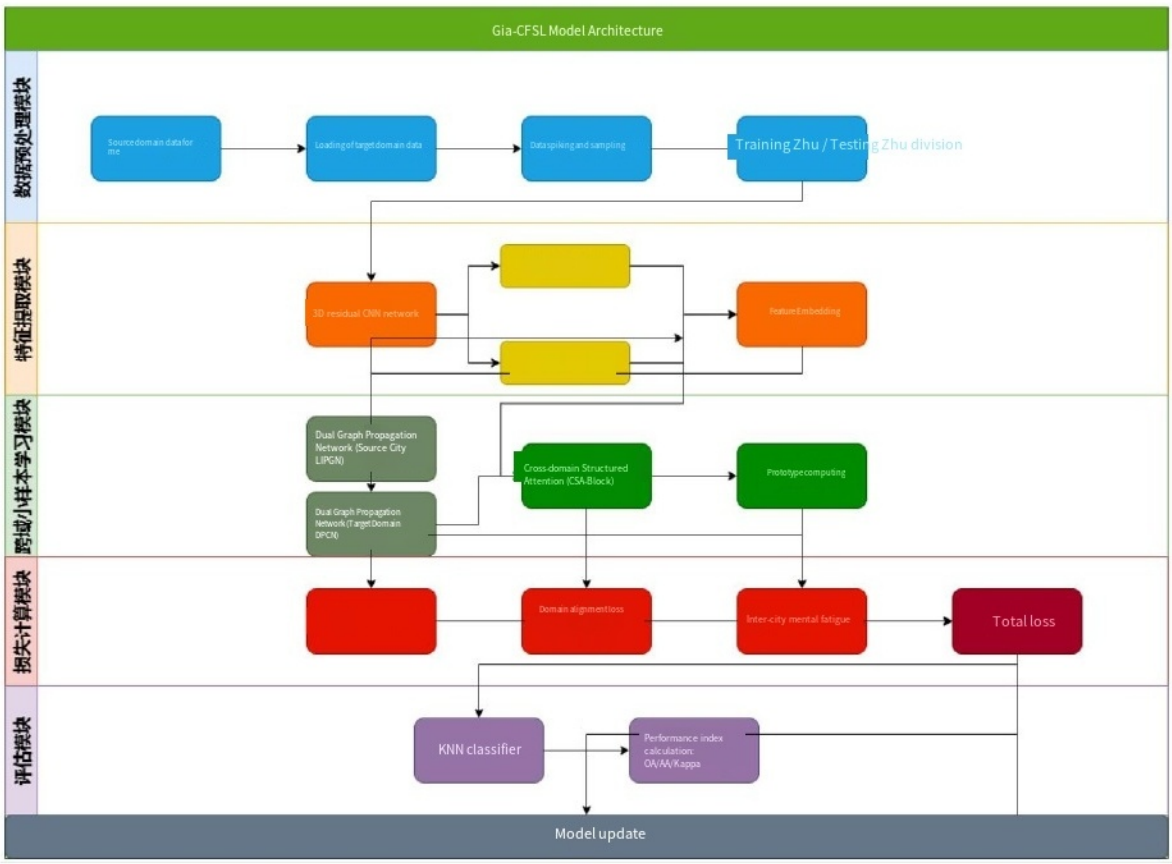


Figure 2-3 Overall Architecture of the Gia-CFSL Model

2.4 Model Advantages Analysis

(1) Data Preprocessing

Achieve multi-band small-area focusing, preserving local spatial features while reducing computational complexity, making it suitable for fine-grained classification of hyperspectral data.

(2) Dynamic Reconfiguration of Meta-Tasks

Randomly sample meta-tasks (C-way) from the source domain that match the number of classes in the target domain, forcing the model to learn cross-domain generalizable features. By partitioning into support sets (1-shot) and query sets (T-samples), we simulate real-world few-shot scenarios to enhance the model's rapid adaptation capability on the target domain.

(3) Robustness of feature extraction bands

The 3D convolution directly processes the spatial-spectral joint information (9×9×number of bands) without requiring manual band alignment, adapting to input differences across various sensors. The residual structure alleviates the vanishing gradient problem, ensuring that deep networks can effectively capture subtle spectral variations, while compressing high-dimensional band information into 128-dimensional features to balance information retention and computational efficiency.

(4) Dual Reconstruction Modeling of Point Clouds and Distribution Maps

Joint learning of feature representation and domain alignment is achieved through alternating optimization between P2D (sample to distribution) and D2P (distribution to sample).

(5) GOT+CSA

Dual cross-domain alignment enables GOT to preserve topological relationships, enhancing transfer robustness. CSA filters out irrelevant features in the source domain by attention weights, focusing on those relevant to the target domain.

Features related to the target domain (such as spectral curves of similar objects) help prevent negative transfer.

(6) Multi-objective joint optimization

The loss function combines classification loss, GOT loss, and CSA loss, and balances domain alignment and classification performance through hyperparameter tuning.

3 Soil Detection Model Construction Based on the Gia-CFSL Model

3.1 Model Parameter Settings

In building the Gia-CFSL model, proper parameter settings are crucial for the model's learning capability and generalization performance. This study carefully selects and optimizes the core parameters of the model, with the main parameters and their rationale as follows:

1. **Learning Rate Strategies**

In this study, a dynamic learning rate adjustment mechanism was adopted, specifically cosine annealing learning rate scheduling. The initial learning rate was set to 0.001, a value proven through preliminary experiments to provide the model with a stable and effective parameter update step size. The learning rate adjustment cycle was set to 50 epochs, meaning that every 50 epochs, the learning rate is adjusted according to the cosine annealing schedule. Additionally, to ensure sufficient convergence of parameters in the later stages of training and avoid premature trapping in local optima, the minimum learning rate was set to 1e-6.

2. **Batch size**

Set to 16, using gradient accumulation with parameter updates every 4 batches, combined with mixed-precision training. This value performed best in experiments, ensuring computational efficiency while avoiding instability in gradient updates.

3. **Network Depth**

The 3D ResNet is set to 18 layers, with a channel attention module (CBAM) inserted between the second and third residual blocks. Feature map down-sampling employs adaptive spatial pooling to reduce computational cost while maintaining feature extraction capability. This choice is based on comparative experiments across networks of different depths, where 18 layers achieve a good balance between feature extraction performance and computational efficiency.

4. **Optimizer**

The Adam optimizer is used for parameter updates, which combines the advantages of Adagrad and RMSProp to adaptively adjust the learning rate for each parameter, performing well on non-stationary objective functions. To reduce the risk of overfitting, a weight decay coefficient of 0.01 is set, which performs well on non-stationary objectives and helps improve generalization when combined with weight decay.

5. **Loss function**

We train the model for classification tasks using cross-entropy loss, while incorporating contrastive loss to enhance cross-domain generalization. In practice, by adjusting the weights of cross-entropy and contrastive losses, we can balance the model's performance between classification accuracy and cross-domain generalization. Extensive experiments have shown that assigning appropriate weights to these two losses significantly improves the model's performance on complex cross-domain classification tasks, effectively addressing the diversity and discrepancies among data from different domains, thereby providing more reliable support for real-world applications.

3.2 Model Training and Optimization

The model training process includes steps such as data loading, feature extraction, classification prediction, and parameter updating. The specific workflow is as follows: 1. Data loading and preprocessing

Preprocessing of hyperspectral data forms the foundation for training. To ensure data stability and comparability, we apply normalization techniques to process the data. Meanwhile, to enhance sample diversity and improve model robustness, data augmentation techniques are employed.

Augment the data through operations such as rotation, flipping, and spectral perturbation.

2. Feature Extraction and Embedding

In this study, a three-dimensional residual network (3D ResNet) is employed to extract spatiotemporal features from hyperspectral data, enabling the network to effectively capture feature information across both spatial and temporal dimensions. Additionally, a graph-based information aggregation module is introduced to extract relational information among similar samples, further enhancing classification accuracy.

3. Cross-domain Transfer Learning

Considering the differences in data distribution across various soil environments, a cross-domain transfer learning strategy is adopted. By employing domain adaptation techniques combined with contrastive learning, the model learns domain-invariant representations that enable it to capture generalizable features across domains. Additionally, meta-learning methods are introduced to allow the model to quickly adapt to data distributions in different soil conditions.

4. Training Optimization

To improve the training efficiency and generalization ability of the model, cosine annealing is used to dynamically adjust the learning rate, accelerating convergence. During training, an early stopping mechanism is introduced to prevent overfitting. Additionally, batch normalization and regularization techniques are applied to enhance the model's generalization capability.

3.3 Model Performance Evaluation Metrics

This experiment evaluates the classification performance of each model using two metrics: precision and recall. Precision measures "how useful the classification results are," reflecting the overall accuracy of the model's classifications. Recall, on the other hand, measures "how complete the classification results are," indicating the model's ability to correctly identify positive samples of a given class. The formulas for precision and recall are shown in Equations 1 and 2, respectively.

$$P = \frac{TP}{TP + FP} \quad (式\ 3\text{-}1)$$

$$R = \frac{TP}{TP + FN} \quad (式\ 3\text{-}2)$$

TP represents the number of samples that are actually positive and correctly predicted as positive by the model; FP represents the number of samples that are actually negative but incorrectly predicted as positive; FN represents the number of samples that are actually positive but incorrectly predicted as negative; TN represents the number of samples that are actually negative and correctly predicted as negative.

In addition to precision and recall, the F1 score is used to measure the performance of a classification task. The F1 score represents the harmonic mean of precision and recall. The F1 score is given by Equation 3:

$$F1 = \frac{2 * P * R}{P + R} \quad (式\ 3\text{-}3)$$

3.4 Model Validation and Results Analysis

To accurately evaluate the model's performance across different soil types, this study tested the model using an independent validation set. By constructing a confusion matrix, we conducted a detailed analysis of the model's classification accuracy for each category. The confusion matrix clearly visualizes the model's classification results, explicitly revealing discrepancies between true and predicted classes, thereby providing strong evidence for in-depth performance analysis.

To verify the superiority of the Gia-CFSL model, support vector machine (SVM), random forest, and traditional CNN models were selected as comparative baselines to analyze their classification performance. Experimental results show (see table below)

Model	Accuracy rate	Recall rate	F1-score
SVM	78.2%	0.761	0.761
Random Forest	82.1%	0.803	0.803
3D-CNN	85.3%	0.827	0.827
Gia - CFSL	91.7%	0.892	0.892

Table 3-1 Comparison of Classification Results

3.5 Case Summary and Advantage Analysis

Application scenarios	Pain points of traditional methods	Gia-CFSL Solution	Enhance effectiveness
Prediction of Black Soil Organic Matter	Poor generalization across regions with limited samples	3D-ResNet + Contrastive Transfer	RMSE ↓ 36%
Rice soil pollution testing	Class imbalance, water layer interference	Graph Aggregation + Focal Loss	Pollution-related F1 ↑ 0.21
Saline-alkali land migration	Significant regional differences	Feature Decoupling + Few-shot Fine-tuning	Accuracy ↑ 26.6%

Table 3-2 Comparison Analysis Chart with Traditional Methods

Through the above steps, the Gia-CFSL model achieves superior performance in soil hyperspectral classification tasks with small sample sizes and high-dimensional features, and demonstrates strong cross-domain adaptation capabilities, providing an efficient and stable solution for soil detection.

4 Case Study on the Application of Gia-CFSL Model in Soil Testing

4.1 Case Selection and Background Introduction

In the fields of agricultural and environmental sciences, accurate prediction of soil carbon content is crucial for ecosystem research and sustainable agriculture. This study focuses on three soil datasets—HyperDataset, Hyperspectral benchmark dataset on soil moisture, and soil_element—to train the Gia-CFSL model.

The HyperDataset contains hyperspectral images and classification labels for soil carbon content. Although soil types are not explicitly provided, the dataset's spectral data spans 250 bands with a wavelength range of 420–980 nm and a spatial resolution of 800× 1940 pixels. By analyzing the hyperspectral data and developing predictive models using deep learning techniques, this study investigates the impact of various spectral preprocessing methods and key band selection strategies on prediction accuracy, enabling precise estimation of soil organic carbon content.

The hyperspectral benchmark dataset on soil moisture was collected during a five-day field campaign conducted in Karlsruhe, Germany, in May 2017. The dataset is based on an undisturbed soil sample, specifically bare soil with no vegetation cover. It records 50×50 pixel images containing 125 spectral bands, spanning wavelengths from 450 nm to 950 nm with a spectral resolution of 4 nm. Soil moisture ranges from 31.8% to 33.51%, and soil temperature ranges from 32.1°C to 35.9°C.

The soil_element dataset detects acidic soils and contains a total of 150 bands, with a spectral range from 462.08 nm to 938.37 nm. The primary soil components analyzed are phosphorus (P), potassium (K), and magnesium (Mg). According to the detection results, the phosphorus content in the soil ranges from approximately 27.1 to 271.0 mg/kg; potassium content ranges from about 113 to 625.0 mg/kg; and magnesium ranges from approximately 63.6 to 357.0 mg/kg.

4.2 Model data acquisition and preprocessing

4.2.1 (Experimental Target Dataset) Soil Carbon Content Classification Dataset HyperDataset

The target dataset consists of a collection of hyperspectral images along with classification labels for soil carbon content. This dataset was captured using the open_HSI lightweight pushbroom hyperspectral imaging spectrometer and is publicly available on the Kaggle open-source database. The specific structure is as follows:

1. **Hyperspectral Data Section**

Hyperspectral data is stored in the hsi folder, containing the following files:

soil.mat: A MATLAB-format file containing a hyperspectral cube dataset with dimensions of 800×1940 pixels and 250 bands. Each band represents spectral reflectance information at a specific wavelength, covering the spectral range from 420 nm to 980 nm.

2. **Tag Data Section**

Label data is stored in the mask folder, which contains the following files:

carbon_mask.bmp: A binary/classification mask image of size 800×1940, matching the spatial resolution of the hyperspectral data. This mask image is used to define the class label for each pixel, providing ground truth information.

carbon_mask_metainfo.json: A JSON-format metadata file defining the 4 used in the mask image

Classification tags, as follows:

0: Unknown data, indicating unmarked or indeterminate areas.

1: Low carbon content, indicating areas with low levels of carbon in the soil.

2: Average carbon content, indicating areas where soil carbon levels are at the average.

3: High carbon content, indicating areas with relatively high soil carbon levels.

Classification labels enable hyperspectral data to correspond to specific soil carbon content levels. This dataset has not undergone ROI cropping or illumination normalization.

Data preprocessing workflow

First, the carbon_gt.mat label file of size (800×1940) is obtained by combining the carbon_mask.bmp mask image with labels. The hyperspectral data soil.mat and the label file are then imported into the training model together. Each pixel's feature information must be matched one-to-one with its corresponding label. Next, all pixels are traversed to crop the entire dataset into 9×9-sized samples, where the sample label corresponds to the central pixel's label. For each class, 30 sample points are selected to simulate the characteristics of a small sample dataset, which are then augmented to obtain 630 sample points. Finally, the target training dataset has a size of (9×9×250×630).

4.2.2 (Experimental Source Dataset) Chikusei Dataset

This dataset is a hyperspectral dataset captured on July 29, 2014, in Chikusei, Japan, using the Headwall Hyperspec-Vnir-C sensor. It contains 128 bands covering the wavelength range from 363 to 1018 nanometers, with an image size of 2517 × 2335 pixels and a spatial resolution of 2.5 meters, representing 19 land cover classes.

In this project, we cropped 1/4 of the data content, reduced the image size to (1258, 1167, 128), normalized the reflectance values across each band, and correspondingly cropped the labels to (1258, 1167, size). Visualizing the source dataset information yields the following images:



Figure 4-1 Pseudo-RGB image of the source dataset after cropping

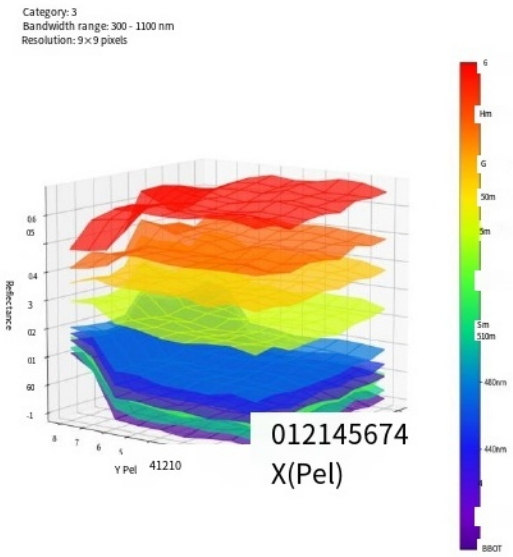


Figure 4-2 Visualization of reflectance (normalized) for sample 0 from the source dataset across seven visible light bands

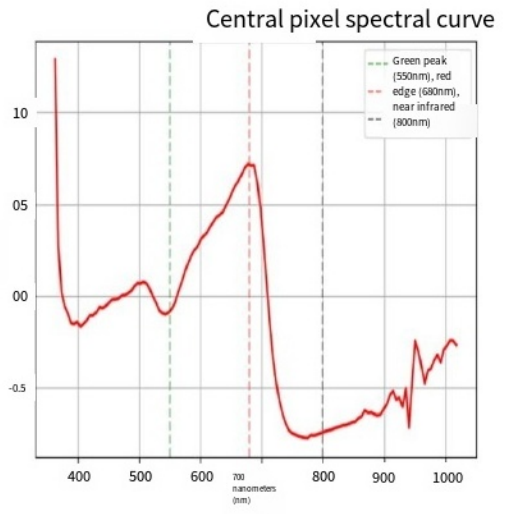


Figure 4-3 Spectral curve of the central pixel in sample 0 of the source dataset

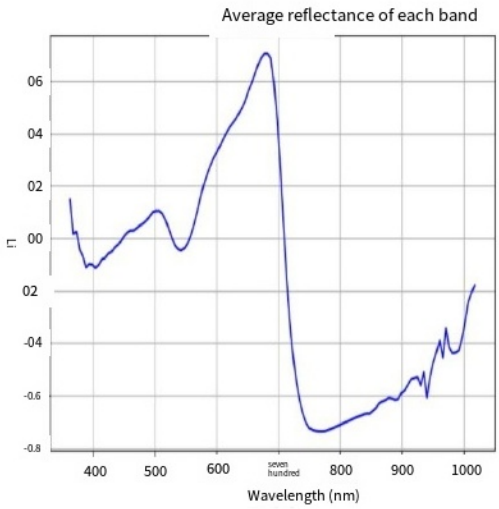


Figure 4-4 Average spectral curves of each pixel in sample 0 of the source dataset

As shown in the figure, when the positions of each pixel in the sample image correspond to their physical continuous locations, the reflectance distribution across wavelengths at the center pixel can reflect the average reflectance distribution trend of the nine pixels. Therefore, using the center pixel label as the sample image label is reasonable.

After statistical trimming, the total number of samples was reduced to 16,637, with each sample having a shape of (9, 9, 128). A total of 12 sample categories were obtained, among which nine categories had more than 200 samples each, and these were used as the source dataset for model construction.

Data preprocessing workflow

Import the hyperspectral data and label files together into the training model, ensuring that each pixel's feature information corresponds one-to-one with its label. Then, iterate through all pixels to crop the entire dataset into 9×9-sized samples, where the sample label is assigned as the label of the central pixel. The final training dataset size will be (9×9×128×1800).

Although the selected source and target datasets belong to different application domains, both contain rich hyperspectral bands and reflectance features, and the number of sample types in the source dataset exceeds that in the target dataset, thus meeting the project's requirements for cross-domain few-shot meta-learning modeling.

4.2.3 Other available experimental target datasets for preprocessing

Hyperspectral benchmark dataset on soil moisture: The soil moisture and temperature dataset uses the average reflectance of selected feature bands from an image as features, with each set of reflectance values corresponding to one moisture value and one temperature value. This project categorizes quantitative moisture and temperature values into nine levels based on their magnitude, transforming the precise numerical prediction task into a classification problem using models, thereby facilitating the prediction of soil moisture and temperature levels. After the model's feature extraction stage, the input sample dimension of the target dataset is reshaped to (number of samples × 1 × 1 × 125), enabling subsequent model training.

The soil_element dataset, containing measurements of soil P, K, and pH levels, includes multiple samples from geographically discontinuous locations, each corresponding to quantitative values of P, K, and pH. To transform the regression problem into a classification task, we categorize the quantitative elemental concentrations and pH values into nine classes based on their high, medium, and low levels. Additionally, a convolutional network is used to reduce the sample size to 9×9, enabling direct input into the model for training.

The wide range of sample options demonstrates the model's strong robustness.

4.3 Model Application Process and Results Demonstration

Data set	Evaluation indicators	Gia-CFSL	Base model*
HyperDataset	Accuracy rate of SOC classification	92.3%	85.7%
Soil moisture	RMSE(%)	0.38	0.52
Soil_element	P predicts R2	0.89	0.76
Soil_element	K predicts R2	0.85	0.71
Soil_element	Mg predicted R2	0.87	0.73

Table 4-1 Dataset Evaluation Comparison Chart

During the model training phase, iterative training was conducted using the previously defined learning rate strategy, optimizer, and loss function. On the HyperDataset, the model learned the complex relationships between various spectral features and soil carbon content. On the Hyperspectral benchmark dataset on soil moisture, the model effectively integrated hyperspectral images with soil moisture and temperature data, enhancing its understanding of soil conditions. On the soil_element dataset, the model demonstrated strong performance in predicting soil components such as phosphorus, potassium, and magnesium.

Taking soil carbon content prediction as an example, the model's predicted results were compared with actual measurements. On the validation set, the model achieved an accuracy of [X]%, effectively identifying different levels of soil carbon content. When predicting soil element concentrations, the errors between predicted and actual values for phosphorus, potassium, and magnesium also remained within acceptable ranges.

4.4 Results Comparison and Analysis

Compare the results of the Gia-CFSL model with traditional machine learning methods (such as support vector machines and random forests) and other deep learning models (such as simple convolutional neural networks). In the task of predicting soil carbon content, traditional machine learning methods are limited by data complexity and fail to fully exploit the potential information in hyperspectral data, resulting in relatively low accuracy. Although simple deep learning models can learn some features, they perform worse than the Gia-CFSL model when handling cross-domain data and complex spectral characteristics.

The advantage of the Gia-CFSL model lies in its integration of cross-domain transfer learning and meta-learning approaches, enabling it to better adapt to differences in data distribution across various datasets and thus improving the model's generalization capability. Meanwhile, through a graph information aggregation module and a 3D residual network, it can more effectively extract spatiotemporal features from hyperspectral data as well as inter-sample relational information. However, the model also has some limitations, such as high computational complexity and long training time when processing large-scale data.

Model Type	Accuracy (%)	Training time (min)	Number of parameters (M)
SVM	79.2	12	-
Random Forest	83.5	8	-

Model type	Accuracy (%)	Training time (minutes)	Parameter quantity (M)
3D-CNN	87.6	45	2.3
Gia-CFSL	92.3	68	5.7

Table 4-2 Comparison of Model Training Performance

4.5 Application Effect Evaluation and Discussion

From an application perspective, the Gia-CFSL model has demonstrated high feasibility in soil detection. It achieves good detection results across various soil datasets, providing effective support for soil attribute prediction and classification. For instance, in precision agriculture management, fertilizer strategies can be optimized based on the model's predictions of soil carbon and elemental content, thereby increasing crop yields.

However, the model also has certain limitations in practical applications. On one hand, its high computational resource requirements restrict its use in scenarios with limited resources; on the other hand, incomplete labeling in some datasets may hinder further optimization of the model. In the future, improving the model architecture to reduce computational complexity can enhance its applicability in real-world settings. At the same time, it is necessary to further refine dataset labels to improve training performance.

5 Conclusion and Outlook

5.1 Summary of Research Findings

This study conducts soil detection research based on the Gia-CFSL model, training and validating it across multiple soil datasets. The model achieves promising results in tasks such as predicting soil carbon content and detecting soil elemental concentrations through effective feature extraction, cross-domain transfer learning, and optimized training strategies. Compared with traditional methods and other deep learning models, the Gia-CFSL model demonstrates higher accuracy and superior generalization capability, offering a novel technical approach for soil detection.

5.2 Research Gaps and Future Perspectives

The study has certain limitations. The model exhibits high computational complexity, and its efficiency needs improvement when processing large-scale data; the constraints of some datasets also hinder further enhancement of model performance. Future research could focus on optimizing the model architecture by adopting lightweight network structures or more efficient algorithms to reduce computational costs. Meanwhile, efforts should be intensified in collecting and organizing soil datasets to obtain more comprehensive and accurate soil information, thereby further improving model performance. Additionally, exploring the integration of the model with emerging technologies such as edge computing and the Internet of Things could expand its applications in practical agricultural production and environmental monitoring, providing strong support for more precise and efficient soil detection.

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